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SYSTEMATIC REVIEWS AND META-ANALYSES

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Global Epidemiology of Gallstones in the 21st Century: A Systematic Review and Meta-Analysis

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BACKGROUND & AIMS: **Gallstones are common and associated with substantial health and economic burden. We aimed to comprehensively evaluate the prevalence and incidence of gallstones in the 21st century.**

METHODS: **We systematically searched PubMed and Embase to identify studies reporting the prevalence and/or incidence of gallstones between January 1, 2000, and November 18, 2023. Pooled prevalence and incidence were calculated using DerSimonian and Laird's random-effects model. We performed subgroup analyses and meta-regression based on age, sex, geographic location, population setting, and modality of detection to examine sources of heterogeneity.**

RESULTS: **Based on 115 studies with 32,610,568 participants, the pooled prevalence of gallstones was 6.1% (95% CI, 5.6–6.5). Prevalence was higher in females vs males (7.6% vs 5.4%), in South America vs Asia (11.2% vs 5.1%), in upper-middle-income countries vs high-income countries (8.9% vs 4.0%), and with advancing age. On sensitivity analysis of population-based studies, the prevalence of gallstones was 5.5% (95% CI, 4.1–7.4; n = 44 studies), and when limiting subgroup analysis to imaging-based detection modalities, the prevalence was 6.7% (95% CI, 6.1–7.3; n = 101 studies). Prevalence has been stable over the past 20 years. Based on 12 studies, the incidence of gallstones was 0.47 per 100 person-years (95% CI, 0.37–0.51), without differences between males and females, and with increasing incidence in more recent studies.**

CONCLUSIONS: **Globally, 6% of the population have gallstones, with higher rates in females and in South America. The incidence of gallstones may be increasing. Our findings call for prioritizing research on the prevention of gallstones.**

Keywords: Gallstones; Prevalence; Incidence; Meta-Analysis.

Gallstones are a common gastrointestinal disorder and a leading cause of hospital admissions, associated with substantial health and economic burden.[1,2] Annually, more than 50,000 cholecystectomies are performed in the United Kingdom, and that number in the United States reached approximately 800,000, costing nearly 6.5 billion dollars each year.[1,3–6] Additionally, recent studies have suggested gallstones may predispose patients to other conditions, such as cardiovascular disease,[7] cancer,[8] and even higher mortality,[9] making it a major health concern worldwide.

Abbreviation used in this paper: CT, computed tomography.

Most current article

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Studying the frequency of gallstones can help understand its distribution and provide clues for developing effective policies for prevention and management.[2] Prevalence and incidence have been reported to be approximately 15% and 0.6% per year, respectively.[10–12] However, owing to variability in detection modalities,[13,14] geographic background,[15,16] and population stratification,[17,18] epidemiologic estimates have been inconsistent.

Although previous reports have tried to summarize the prevalence and incidence of gallstones,[1,5,10–12] the studies covered in these reviews were not comprehensive. In addition, the differences between factors that are of clinical importance, such as detection modalities and gallstone disease subtypes, have not been evaluated thoroughly. With the ongoing increase of metabolic dysfunction and unhealthy lifestyle and dietary patterns, the burden of gallstones was expected to increase in parallel.[19,20] However, most of the studies included in previous reviews were published before 2000.[5,10–12] Thus, a comprehensive study based on the latest evidence was warranted to understand the current epidemiology of gallstones worldwide.

The purpose of this study was to conduct a systematic review and meta-analysis to comprehensively evaluate the global prevalence and incidence distribution of gallstones after the year 2000. We also aimed to distinguish the differences in gallstone frequency by different geographic background, detection modalities, publication year, age, sex, and source of participants to inform health care policy and prioritization.

What You Need to Know

Background

Several studies have investigated global gallstone disease frequency with inconsistent results. We aimed to comprehensively evaluate the prevalence and incidence of gallstone disease in the 21st century.

Findings

There was a high prevalence and incidence of gallstone disease around the globe. Pooled gallstone disease prevalence was approximately 6%, and there was geographic variability.

Implications for patient care

The results quantify the global burden of gallstone disease and could provide valuable guidance for allocating health care resources appropriately.

Materials and Methods

Search Strategy and Selection Criteria

We conducted this systematic review and meta-analysis according to the Meta-Analysis of Observational Studies in Epidemiology guidelines.[21] The primary outcomes were the prevalence and incidence of gallstones, which was defined as the presence of stones in the gallbladder and bile ducts, or absence of the gallbladder as a result of cholecystectomy.[22,23] We searched PubMed and Embase databases to identify relevant studies published between January 1, 2000, and November 18, 2023. The medical subject heading terms including “prevalence” OR “incidence” OR “morbidity” OR “frequency” AND “gallstone” OR “gallstone disease” OR “cholelithiasis” OR “gallbladder stone” OR “gallbladder calculus” OR “biliary tract stone” OR “bile duct stone” OR “choledocholithiasis” OR “cholecystectomy” were used. Listed references also were checked manually to identify relevant reports.

The inclusion criteria were as follows: (1) cross-sectional or cohort-designed studies that investigated the prevalence or incidence of gallstones in the general population; (2) in case-control studies, only healthy controls originating from the general population (not matched to the case group) were included; (3) cross-sectional research should provide the number of cases and total population; (4) longitudinal studies should provide the number of cases and follow-up period or person-years, and studies focusing on gallbladder diseases (defined as gallstones, cholecystectomy, or cholecystitis)[24] or on populations with specific morbidities such as cirrhosis or gastrectomy were excluded; and (5) studies focused primarily on evaluations conducted before 2000 also were excluded. When multiple studies focused on the same population, we only included the most comprehensive study with the largest sample size, or the study with the most extended follow-up period.

Data Extraction

Two authors independently extracted relevant information using a predesigned collection form, including PubMed Identifer, first author, publication year, geographic background, study period, source of the participants (community/health checkup population/factory or employee-based), study outcome (gallstone disease/gallstones/cholecystectomy), detection modalities (autopsy/questionnaire-acquired/medical record/ultrasonography or computed tomography of the abdomen), follow-up year, person-years, total population sample size, number of male and female participants, mean age, and number of patients with gallstones. Any discrepancies were resolved by consensus.

Statistical Analysis

Heterogeneities between studies were evaluated by the Cochran’s Q test and I^2 was used for assessing

variance.[25,26] We also conducted meta-regression to track the potential source of heterogeneity. Because of the relatively higher level of heterogeneity across studies, the DerSimonian and Laird random-effects model was performed to calculate the pooled prevalence and incidence of gallstones.[27] The distribution of the pooled prevalence and incidence in each country was combined further and visualized on a world map. Subgroup analyses were conducted stratifying by geographic background, gender, detection modalities, source of participants, and publication year. Furthermore, we also categorized all sources of studies into 4 groups based on the World Bank's income level classifications, namely the high-income countries, upper-middle-income countries, lower-middle-income countries, and low-income countries according to the World Bank classification of economy.[28] Because income levels in different countries fluctuate over time (Supplementary Figure 1), we defined the income level of the study as the income level of the country when the first participant enrolled in the study. If the number of studies in a certain subgroup was fewer than 2, a meta-analysis was not conducted. Heterogeneities between subgroups also were evaluated by the Cochran's Q test. Because of distinct classification of age groups in different studies, we used 2 approaches to combine the age-specific gallstone frequency (0, 1–20, 21–30, 31–40, 41–50, 51–60, 61–70, and >70 years age groups, and 0, 1–34, 35–44, 45–54, 55–64, 65–74, and >74 years age groups). We plotted a line chart to display the trend of pooled age-specific gallstone frequency. We also performed a sensitivity meta-analysis based exclusively on the community population to measure the robustness and reliability of the pooled results in population-based studies. If each data set did not show any obvious influence on the pooled estimates, it then confirmed the findings in the direction and magnitude of the present meta-analysis.[29] Publication bias was tested by visualizing the asymmetry of a funnel plot and the Egger and Begg regression tests.[30,31] The pooled prevalence and incidence of gallstones were synthesized using Comprehensive Meta-Analysis version 3.7 (Biostat, Englewood, NJ). The world map was plotted using Tableau Desktop Personal version 10.3 (Tableau, Seattle, WA).[32] Statistically significant levels were 2-tailed and set at $P < .05$. This study was registered with the International Prospective Register of Systematic Reviews (registration number: CRD42020191822).

Results

Characteristics of the Included Studies

Initially, we identified 31,351 records from PubMed and Embase databases. After excluding duplicates and screening titles, abstracts, and full texts, 207 records met the eligibility criteria, and 23 studies were retrieved manually. Fifty-two studies were further eliminated because they focused on the same population but with relatively smaller sample sizes or shorter follow-up periods, and 51 studies also were excluded because they focused primarily on evaluations conducted before the year 2000. Finally, 127 studies incorporating approximately 38 million participants were included (Figure 1).

Characteristics of all included studies are reported in Supplementary Table 1. Of the included studies, 115 reported the prevalence of gallstones (4 studies were included only for subgroup analysis). According to country, 81 studies were conducted in Asia, followed by Europe (N = 12), South America (N = 9), North America (N = 7), Africa (N = 5), and Oceania (N = 1). A total of 99.4% of participants were recruited from hospitals and communities, and 52.4% of them were diagnosed by ultrasonography or computed tomography (CT).

A total of 12 cohort studies were included for evaluating the incidence of gallstones (9 in Asia, 2 in Europe, and 1 in America), approximately 58.6% of patients were recruited from the community, and 39.9% were recruited from hospitals. Regarding detection modalities, 6.7% of the studies confirmed the presence of gallstones in the study subjects by medical record, and 93.2% were based on ultrasonography or CT.

Prevalence of Gallstones

A total of 115 studies evaluating the prevalence of gallstones were included, and the combined prevalence was calculated as 6.1% (95% CI, 5.6%–6.5%) among 32,610,568 participants (Table 1 and Supplementary Figure 2). Sensitivity analysis focusing exclusively on the community population showed a slightly lower pooled prevalence of gallstones (5.5%; 95% CI, 4.1%–7.4%) (Table 2). The distribution of gallstone prevalence among different countries ranged from 1.93% in India to 25.8% in Mexico (Figure 2*A*). At the continent level, South America had the highest pooled prevalence at 11.2% (95% CI, 7.5%–11.3%), followed by North America (8.1%; 95% CI, 3.0%–20.2%), Africa (6.6%; 95% CI, 3.6%–11.9%), and Europe (6.4%; 95% CI, 4.2%–9.7%), all with significant difference ($P < .001$). Moreover, a difference was found in pooled gallstone prevalence among Europe, Africa, North America, South America, and Asia (pooled prevalence, 5.1%; 95% CI, 4.6%–5.7%) (P for difference = .005) (Table 1). Furthermore, based on World Bank classification of economy, the pooled prevalence was higher in low-income countries (8.9%; 95% CI, 5.2%–14.7%) and upper-middle income countries (8.9%; 95% CI, 7.6%–10.5%), followed by lower-middle income countries (5.3%; 95% CI, 4.4%–6.4%) and high-income countries (4.0%; 95% CI, 3.3%–5.0%) (P for difference < .005).

Regarding the participants' characteristics, results showed that the pooled prevalence of gallstones was increased markedly with age, with the highest prevalence reaching 13.0% (95% CI, 11.8%–14.4%) in the

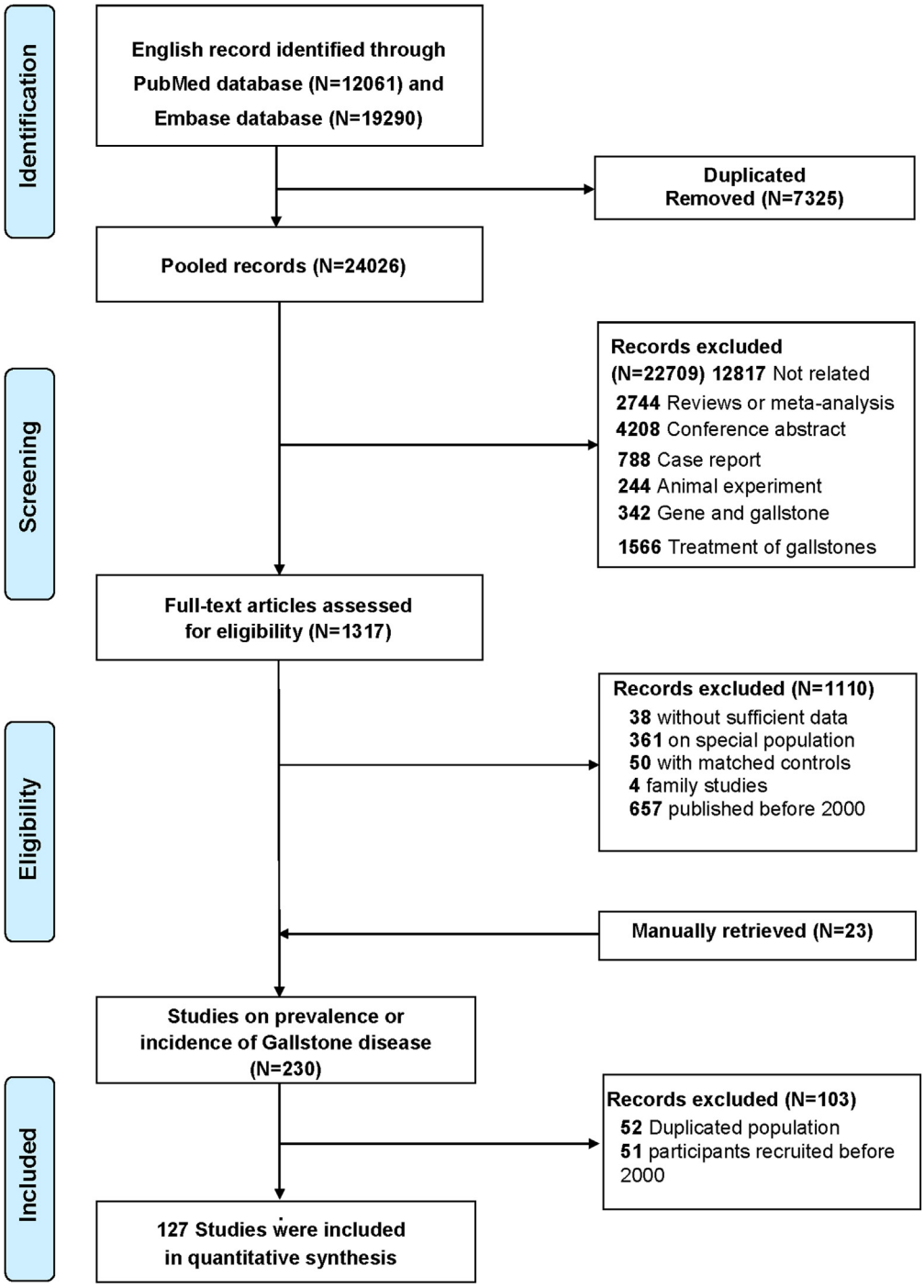

Figure 1. Flowchart of study selection.

>70-year age group and 11.5% (95% CI, 8.1%–24.6%) in the ≥74-year age group (Figure 3*A* and *B*). Females had a significantly higher gallstones prevalence than males (7.6% vs 5.4%; *P* for difference < .001) (Table 1). Concerning different detection modalities, ultrasonography or CT proved to have a significantly higher gallstone detection rate (pooled prevalence, 6.7%; 95% CI, 6.1%–7.3%) than autopsy (4.7%; 95% CI, 2.3%–9.1%), questionnaire (4.3%; 95% CI, 3.0%–6.3%), and medical records (2.2%; 95% CI, 1.2%–4.1%) (*P* for difference <.001). However, the pooled gallstone prevalence did not differ by participant source (*P* for difference = .259) or publication period (*P* for difference = .403). Meta-regression analysis suggested that the mean age, proportion of females, and publication periods partly could explain the heterogeneity in gallstone prevalence between studies (Supplementary Figure 3*A*–*C*).

Incidence of Gallstones

We included 12 studies in the meta-analysis for measurement of gallstone incidence. The pooled incidence was 0.47 (95% CI, 0.37–0.51) per 100 person-years, and sensitivity meta-analysis also confirmed a slight decrease in the crude incidence of gallstones within the community population (Tables 1 and 2).

Table 1. Subgroup Analysis for the Pooled Prevalence and Incidence of Gallstones

Diseases	Records, n	Sample size	I^2, %	Prevalence, %/incidence per 100 PY (95% CI)	*P* for interaction
Combined prevalence	121	32,610,568	99.93	6.1 (5.6–6.5)	
Geographic background					.005
Asia	82	17,729,820	99.93	5.1 (4.6–5.7)	
Africa	5	9562	98.09	6.6 (3.6–11.9)	
Europe	14	20,928	98.72	6.4 (4.2–9.7)	
North America	7	14,832,112	99.96	8.1 (3.0–20.2)	
South America	12	19,673	98.85	11.2 (7.5–11.3)	
Income classifications					<.001
High	31	16,304,915	99.87	4.0 (3.3–5.0)	
Upper-middle	33	14,154,093	99.96	8.9 (7.6–10.5)	
Lower-middle	29	1,544,479	99.66	5.3 (4.4–6.4)	
Low	5	8181	97.95	8.9 (5.2–14.7)	
Sex					<.001
Male	94	15,282,186	99.91	5.4 (4.9–5.9)	
Female	98	16,125,408	99.83	7.6 (7.1–8.2)	
Detection modalities					<.001
Medical record	6	15,746,500	99.97	2.2 (1.2–4.1)	
Questionnaire acquired	8	613,816	99.75	4.3 (3.0–6.3)	
Ultrasonography or CT	101	16,247,825	99.90	6.7 (6.1–7.3)	
Autopsy	3	583	97.17	4.7 (2.3–9.1)	
Source of participants					.259
Community-based	44	1,906,385	99.89	5.5 (4.1–7.4)	
Health checkup population	68	30,495,493	99.90	6.4 (5.9–7.0)	
Factory- or employee-based	6	204,820	99.86	4.2 (2.2–7.8)	
Publication year					.403
2000–2005	14	47,011	99.34	7.6 (4.5–12.5)	
2006–2010	22	140,254	99.37	6.5 (4.8–8.9)	
2011–2015	43	2,661,485	99.89	5.2 (4.3–6.4)	
2016–2020	31	4,033,372	99.96	5.5 (4.0–7.5)	
2021–2023	10	25,727,418	99.96	6.5 (5.6–7.2)	
Combined incidence	12	5,814,164	99.87	0.47 (0.37–0.51)	
Geographic background					<.001
Asia	8	2,123,059	99.88	0.48 (0.37–0.60)	
Europe	2	3,686,521	33.30	0.273 (0.271–0.275)	
Income classifications					<.001
High	8	5,701,046	99.92	0.36 (0.28–0.44)	
Lower-middle	3	108,534	29.77	0.61 (0.60–0.63)	
Sex					.643
Male	7	687,589	99.14	0.45 (0.37–0.53)	
Female	7	603,601	98.44	0.48 (0.41–0.55)	
Detection modalities					.042
Medical record	4	391,980	99.49	0.56 (0.39–0.73)	
Ultrasonography or CT	6	5,417,600	99.94	0.37 (0.27–0.47)	
Source of participants					.200
Community-based	2	3,404,374	94.13	0.41 (0.10–0.80)	
Health checkup population	8	2,320,843	99.93	0.71 (0.56–0.86)	
Publication year					.026
2016–2020	6	1,307,061	99.54	0.47 (0.39–0.55)	
2021–2023	3	494,087	99.96	1.00 (0.54–1.47)	

NOTE. In some instances it was not possible to conduct a corresponding analysis because of insufficient research data, and is indicated as NA.
CT, computed tomography; PY, person-years.

Globally, Japan had the highest pooled gallstone incidence at 6.86 per 1000 person-years. In comparison, the pooled incidence in the United Kingdom was relatively lower than in other countries (2.69 per 1000 person-years) (*P* for difference < .001) (Figure 2*B*). For different continents, the pooled gallstone incidence in Europe and Asia were 0.273 (95% CI, 0.271–0.275) and 0.48 (95% CI, 0.37–0.60) per 1000 person-years, respectively, and with a significant difference (*P* for difference < .001) (Table 1 and Supplementary Figure 4). Additionally, the incidence in lower-middle income countries was relatively lower than that in other countries (2.69 per 1000 person-years) (*P* for difference < .001) (Figure 2*B*).The pooled gallstone incidence increased with the age of participants and peaked at 9.18 (95% CI, 7.87–10.5) per 1000 person-years in the ≥70

Table 2. Subgroup Analysis for the Pooled Prevalence and Incidence of Gallstones Based on Community Population

Diseases	Records, n	Sample size	I^2, %	Prevalence, %/incidence per 100 PY (95% CI)	*P* for interaction
Combined prevalence	44	1,906,385	99.89	5.5 (4.1–7.4)	
Geographic background					.093
Asia	22	1,339,237	99.90	4.7 (3.3–6.6)	
Africa	2	4624	0.000	4.3 (3.8–4.9)	
Europe	11	16,662	98.80	5.1 (3.0–8.8)	
North America	3	531,285	99.99	3.6 (0.2–40.9)	
South America	5	14,259	99.45	13.3 (6.3–25.9)	
Income classifications					.086
High	8	1,262,067	99.94	3.7 (1.4–9.2)	
Upper-middle	8	33,813	99.59	9.6 (5.6–16.0)	
Lower-middle	11	65,505	99.62	4.5 (2.4–8.0)	
Sex					.005
Male	38	988,452	99.60	4.2 (3.2–5.5)	
Female	38	986,120	99.86	7.9 (5.6–11.0)	
Detection modalities					.046
Medical record	2	1,218,479	99.98	0.5 (0.4–4.9)	
Questionnaire acquired	7	569,687	99.66	4.8 (3.4–6.8)	
Ultrasonography or CT	32	109,727	99.29	6.5 (5.0–8.5)	
Publication year					.290
2000–2005	12	46,285	99.45	7.5 (4.4–12.6)	
2006–2010	6	7557	99.18	6.1 (2.3–15.3)	
2011–2015	14	1,082,726	99.89	3.2 (1.7–6.0)	
2016–2020	8	747,059	99.94	5.8 (2.2–14.5)	
2021–2023	3	17,011	97.77	7.6 (3.7–15.1)	
Combined incidence	2	3,404,374	94.13	0.43 (0.10–0.76)	
Sex					.067
Male	2	72,271	89.87	0.46 (0.19–0.72)	
Female	2	21,260	32.70	0.70 (0.61–0.85)	

NOTE. In some instances it was not possible to conduct a corresponding analysis because of insufficient research data, and is indicated as NA.
CT, computed tomography; PY, person-years.

years age group and 29.8 (95% CI, 18.6–41) per 1000 person-years in the 65- to 74-year age group (Figure 3*C* and *D*). The pooled gallstone incidence in males and females was 0.45 (95% CI, 0.37–0.53) and 0.48 (95% CI, 0.41–0.55), respectively, without significant differences (*P* for difference = .643) (Table 1). In subgroup analyses for detection modalities, the medical record (0.56; 95% CI, 0.39–0.73 per 100 person-years) had a higher incidence than ultrasonography or CT (0.37; 95% CI, 0.27–0.47 per 100 person-years) (*P* for difference = .042). No significant difference was found in the source of participants (*P* for difference = .200), and health check-up population studies (0.71; 95% CI, 0.56–0.86 per 100 person-years) reported a higher incidence than the community (0.41; 95% CI, 0.10–0.80 per 100 person-years) (Table 1). Meta-regression analysis suggested that mean age and the proportion of females could not explain the heterogeneity in the gallstone incidence among studies (Supplementary Figure 3*D* and *E*).

Publication Bias

The funnel plot suggested the presence of asymmetry for studies concerning the prevalence of gallstones (Supplementary Figure 5*A*). Moreover, the Begg (*P* = .03) and Egger tests (*P* = .12) indicated a significant publication bias. Additionally, significant publication bias also was detected in studies reporting the incidence of gallstones (Supplementary Figure 5*B*).

Discussion

Overview of the Global Prevalence and Incidence of Gallstones

The present study comprehensively reviewed the global prevalence and incidence of gallstones based on studies published between 2000 and 2023 involving more than 38 million participants. We found that the pooled prevalence and incidence of gallstones were 6.1%, and 0.47 per 100 person-years, respectively, and that gallstone frequency increases with age. We found significant differences according to geographic background, sex, World Bank classification of economy, and detection modality. However, no difference was found in the source of participants and publication periods.

Consistent with previous studies, our results show a significant difference in the pooled gallstone prevalence according to geographic background.[2,5,10] Furthermore, an analysis based on the World Bank's classification of

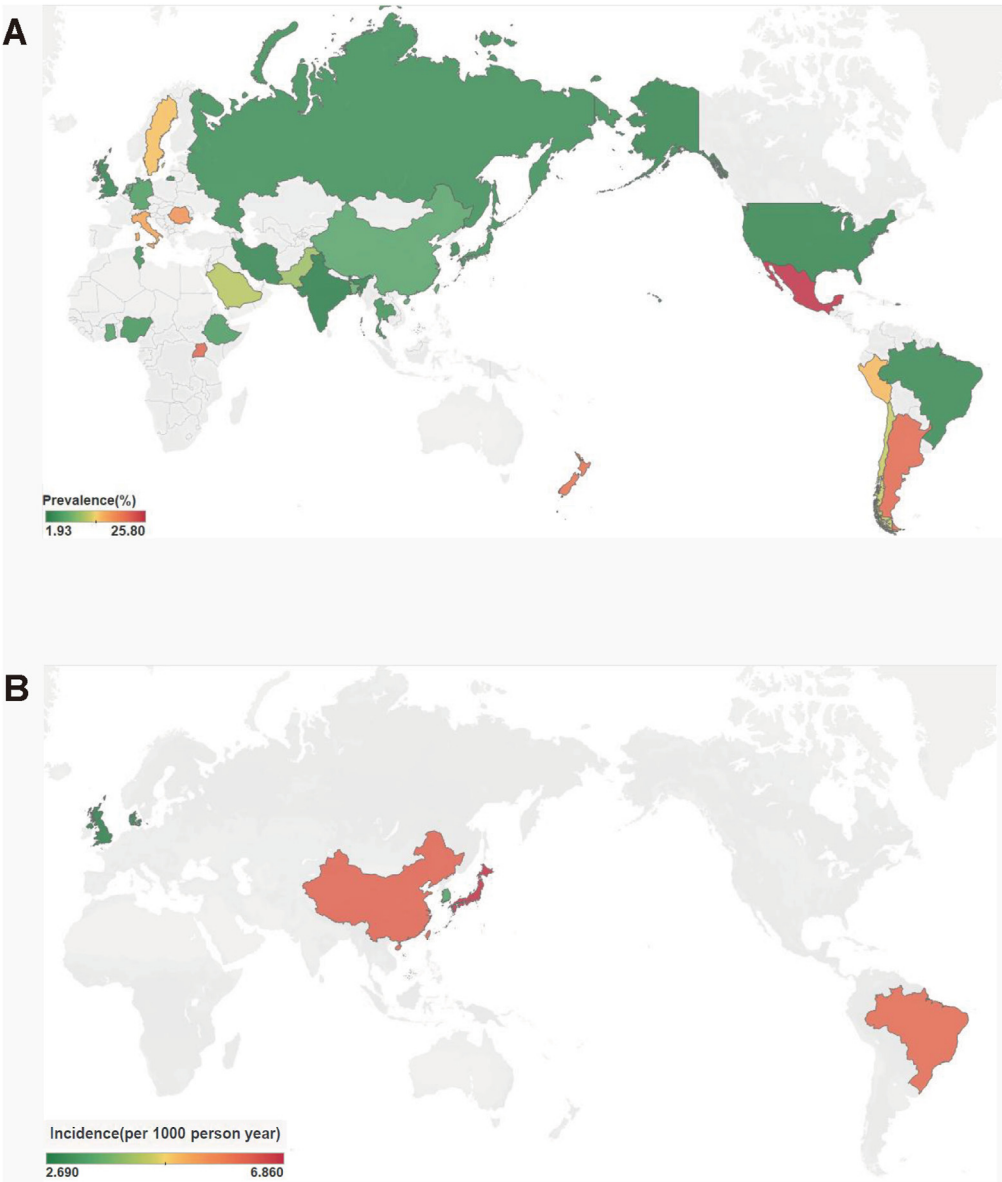

Figure 2. Worldwide distribution of the (*A*) pooled prevalence and (*B*) incidence of gallstones.

national income levels also revealed that both the incidence and prevalence of gallstones are higher in low-income countries. Although the income levels for different countries fluctuate over time, most studies did not show significant fluctuations in income levels in their countries during the recruitment period of study participants. Therefore, the definition of income level for the studies accurately can assess the association between income level and gallstone occurrence risk. Results suggest that different economic development levels partly may explain the differences, which likely are linked to several key gallstone risk factors such as high-calorie diets and obesity.[2,33] Here, ethnicity-related heritability and genetic susceptibility, which were reported to account for approximately 25% of gallstone variance, also may play important roles.[34,35] It is important to note that the prevalence in Oceania was much higher than in other continents. However, because only 1 study was included from Oceania, the results need to be interpreted cautiously. Simultaneously, there is presently a limited amount of research addressing the correlation between economic development and the occurrence of gallstones. Additional studies in this field are warranted for future exploration.

Factors Influencing the Prevalence and Incidence of Gallstones

Females have a significantly higher gallstone prevalence than males. This could be owing to increased risk secondary to hormone replacement therapy,[36] oral contraceptive use,[37] and parity.[38] Female sex hormones also may play a role because it has been reported that estrogen could increase cholesterol secretion, which subsequently may increase the cholesterol saturation of bile.[2] In addition, progesterone could inhibit gallbladder emptying ability, leading to bile stasis.[39] Surprisingly, there was no significant difference in pooled gallstone

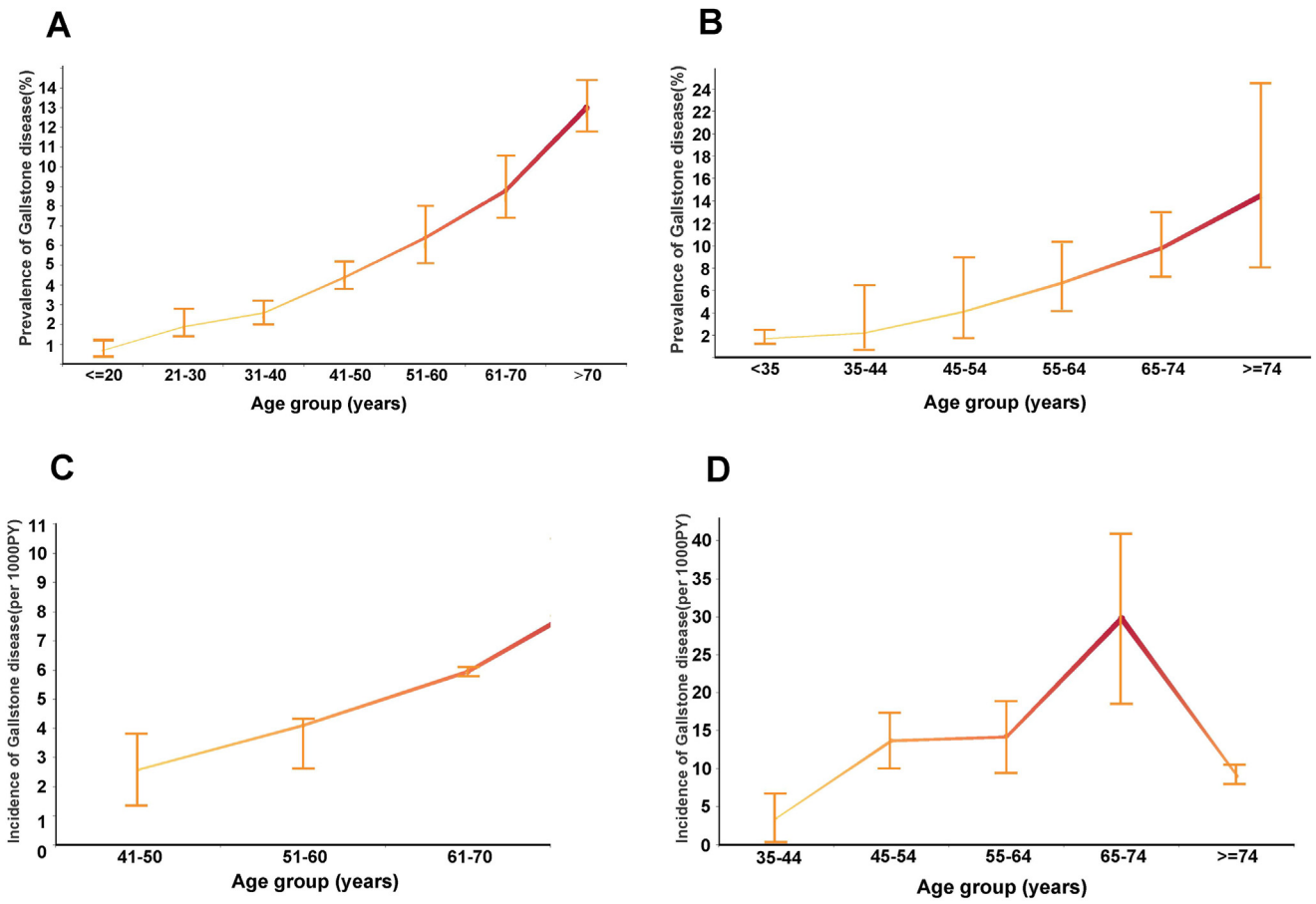

Figure 3. The trend of the (*A* and *B*) pooled prevalence and (*C* and *D*) incidence of gallstone increase with increasing mean age.

incidence between geographic background and sex subgroups, which partly may be owing to smaller effect sizes.

It also was found that different detection modalities heavily influence the prevalence and incidence estimates of gallstones. Although medical history covered all symptomatic patients, asymptomatic gallstone patients were diagnosed mainly by ultrasonography or CT. Because of the relatively lower necropsy rate and the potential selection bias for the inclusion of participants who had passed away and required investigation into the origins of symptoms, autopsy estimates inadequately represent gallstone frequency in the general population. In addition, because of the self-reported nature of these data, misclassification may affect gallstone frequency to a certain extent. As a noninvasive imaging technique that could display a clear image of bile, ultrasonography is recommended to estimate the prevalence and incidence of gallstones in the general population.

Our results also indicate that with progressive publication years, the prevalence of gallstones decreased. This may be related to the increase in the proportion of studies in recent years in which the diagnostic method was based on medical records. Our study also suggests that there was no significant difference in the prevalence and incidence of gallstones between community-based studies, health checkup population-based, and factory- or employee-based studies. However, more studies are needed because of the relatively smaller sample sizes.

As described, gallstone frequency significantly increases with age because older people have more risk factors. Thus, more scrutiny and prophylactic interventions should be targeted to the older population to prevent gallstones. In addition, cohort studies also connect gallstones to other extra-digestive conditions including cardiovascular disease (CVD),[7,40] type 2 diabetes,[41,42] autoimmune disease,[43,44] pancreatitis,[45] cancer,[46–48] kidney stones,[49,50] and even higher mortality rates,[51–53] which could decrease dramatically after treatment, especially surgery.[54] Thus, real-time screening should be conducted for the early identification and timely treatment of gallstone patients. However, only 3 publications were included for analyzing the pooled incidence of gallstones in different age groups, and hence additional data are needed.

The sensitivity analysis indicated that the prevalence and incidence of gallstones were slightly lower in community populations compared with the overall population. The subgroup analysis based on community population showed that gallstone frequency was consistent with the overall population. Thus, the results were conclusive and robust in this study period, which could help explain the current epidemiologic distribution of gallstones.

Limitations and Future Prospects

This study had a few limitations. Although some factors such as geographic background and detection modalities partly could explain the heterogeneities, the overall heterogeneity for all pooled gallstone prevalence and incidence was relatively high, which may have affected the results to some extent. Thus, the findings should be interpreted cautiously, and other potential confounders should be investigated in the future. Although we conducted a broad search in a widely used database, the sample size was small for specific countries and subgroups, limiting the power for assessing gallstone frequency. Additionally, because some of the studies did not clearly define gallstones, we could not distinguish whether these studies focused on gallstones alone or both gallstones and cholecystectomy.

This study provides the summarized estimates for gallstone frequency in the 21st century. The results suggest a high prevalence and incidence of gallstones around the globe, and that gallstone prevalence varies in different countries and continents. Previous studies have suggested that gallstones may have significant downstream consequences and the risk could be diminished in part by treatment. Additionally, the results also demonstrate that the prevalence of gallstones increases with age and in females, which helps clinicians target screening for gallstones in the elderly population and among females. Our findings highlight the disease burden of gallstones and call for screening strategies and prophylactic treatment approaches for high-risk patients.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Clinical Gastroenterology and Hepatology* at www.cghjournal.org, and at http://doi.org/10.1016/j.cgh.2024.01.051.

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Data Availability

All data generated or analyzed during this study are included in this article and its Supplementary material files. Further enquiries can be directed to the corresponding author.